QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

IV SEMESTER

MG4DSEPHYA01– Stellar Astrophysics – Radiative Process, Stellar Structure and Evolution

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **3** marks

1.	Define emissivity. State it's unit.	[K][CO1]
2.	What is spontaneous emission? Define the Einstein A-coefficient.	[U][CO2]
3.	Why is Bremsstrahlung radiation continuous in nature?	[A][CO2]
4.	Write a short note on line broadening.	[U][CO3]
5.	Define Stellar Luminosity.How does it vary with temperature?	[E][CO4]
6.	Write about low mass and high mass star.	[E][CO4]
7.	Write about the thermonuclear fusion in star.	[E][CO4]
8.	Graphically show how does the stellar spectrum of Blue star ,Yellow star and Red star vary.	[E][CO4]
9.	What is Gaint molecular clouds.	[K[CO4]
10.	Explain why white Dwarf are very dense.	[U][CO5]
11.	Draw the evolutionary diagram of final state of star with mass similar to sun.	[A][CO5]
12.	Distinguish between Neutron star and pulsar.	[E][CO5]

[3x10 = 30]

Part B

Short Essay Type Questions
Answer any **4** questions
Each question carries **5** marks

13.	Explain net flux and momentum in radiation field.	[U][CO1]
14.	Explain the physical significance of the 'Classical Electron Radius' (r0) in the context of Thomson scattering and calculate the total scattering cross-section .	[A][CO2]
15.	Explain stellar classification.	[E][CO4]
16.	Explain Virial theorem and the hydrostatic equillibrium in star.	[E][CO4]
17.	Explain the Jeans mass condition ? Discuss its physical significance in star formation.	[A][CO4]
18.	List and briefly explain the stages from protostar to main sequence star.	[K][CO4]

[4x5 = 20]